

Problem Statement

REAL-TIME WATER MANAGEMENT SYSTEM

1. Industry Context

Water is one of the most essential natural resources required for domestic use, agriculture, industries, and public infrastructure. Increasing population, growing water demand, pipeline leakage, inefficient distribution, and irregular consumption create significant challenges for effective water management.

Traditional water management systems often depend on manual inspection and periodic measurements. These methods may not provide sufficient real-time information about water levels, flow rates, consumption, pressure, or leakage. Delayed identification of such problems can result in:

- ◆ Water wastage and resource loss
- ◆ Pipeline leakage and distribution problems
- ◆ Unnecessary water consumption
- ◆ Increased operational and maintenance costs
- ◆ Reduced availability of water resources
- ◆ Delayed response to abnormal conditions
- ◆ Inefficient water distribution

There is a growing need for an intelligent Real-Time Water Management System that continuously monitors water-related parameters, detects abnormal conditions, provides timely alerts, and supports efficient decision-making.

2. Problem Statement

Design and develop an end-to-end Real-Time Water Management System for monitoring and managing water resources using sensors, connected devices, and a centralized monitoring platform.

The system should collect real-time data such as water level, flow rate, water consumption, pressure, water quality parameters, sensor status, and location from water tanks, pipelines, distribution networks, and other water facilities.

Using configurable thresholds and real-time data analysis, the system should automatically detect water leakage, abnormal consumption, low water levels, pressure variations, and other critical conditions. The system should provide timely alerts to users or authorities and maintain historical records for analysis and decision-making.

The solution should provide a user-friendly dashboard for monitoring water conditions, viewing alerts, analyzing consumption patterns, and generating reports. It should support multiple monitoring points while ensuring scalability, reliability, security, maintainability, and continuous availability.

3.Objectives

The system should:

- ◆ To Continuously collect water-related data from sensors and connected devices.
- ◆ To Monitor water levels, flow rates, pressure, consumption, and quality parameters in real time.
- ◆ To Detect water leakage and abnormal water consumption at an early stage.
- ◆ To Identify low water levels and unusual changes in water pressure.
- ◆ To Provide real-time alerts when critical water conditions are detected.
- ◆ To Display monitoring information through a centralized GUI/dashboard.
- ◆ To Store historical water-management data for future analysis.
- ◆ To Provide analytical information to support efficient water-resource management.
- ◆ To Generate reports related to water consumption, leakage, monitoring, and alerts.
- ◆ To Improve the efficiency and reliability of water distribution.
- ◆ To Reduce water wastage through continuous monitoring and early detection.
- ◆ To Apply software engineering principles such as security, scalability, reliability, maintainability, and usability.

Expected Outcome:

1. Develop an end-to-end Real-Time Water Management System for collecting, storing, monitoring, and analyzing water-related information.
2. Provide a GUI/dashboard for monitoring water levels, flow rates, water consumption, pressure, water quality, sensor status, and alerts in real time.
3. Maintain information such as water level, flow rate, water consumption, pressure, water quality, sensor status, location, alerts, monitoring date, and historical records.
4. Enable real-time detection and contextual analysis of water leakage, abnormal consumption, low water levels, pressure variations, and critical conditions.
5. Provide secure and efficient data storage, retrieval, and management for water monitoring information.
6. Allow users to view, analyze, and generate reports related to water usage, leakage detection, monitoring conditions, and alerts.
7. Provide real-time notifications and alerts when predefined water-management thresholds or abnormal conditions are detected.
8. Enable historical data analysis to identify water-consumption patterns, leakage trends, and changes in water-management conditions.
9. Apply key software engineering principles such as security, scalability, maintainability, reliability, usability, and performance.
10. Demonstrate proper requirements analysis, system design, SDLC selection, and evaluation of risks, limitations, and future enhancements.